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July 1, 2022

Mr. Rupert Roy-Clark
Director of Facilities
Amherst-Pelham Regional School District
170 Chestnut Street
Amherst, MA 01002
e.roy-clark@arps.org

Re: Amherst-Pelham Regional Middle School Auditorium Roof Evaluation
170 Chestnut Street
Amherst, MA 01002
Gale JN 839810

Dear Mr. Roy-Clark:

In accordance with our agreement, Gale Associates, Inc. (Gale) performed an evaluation of the auditorium roof at Amherst-Pelham Regional Middle School (APRMS). This roof contains five (5) roof areas – the main auditorium roof and four (4) adjacent lower roofs that exist predominantly over corridor space. The purpose of this letter is to provide Amherst-Pelham Regional School District (APRSD) with an understanding of the existing roof conditions and to provide roof replacement and repair options. It is Gale's understanding that APRSD is considering installing photovoltaic (PV) arrays at the auditorium roof; accordingly, Gale has also been retained to perform a limited structural analysis of the auditorium roof framing under Gale Job No. 839820. The intent of the limited structural analysis is to evaluate the feasibility of installing PV arrays and whether the existing roof framing contains sufficient reserve capacity to support the additional load of the arrays, with consideration for the weight of the new roof assembly. Gale's findings and opinions will be outlined in a forthcoming separate letter report. Note that applicability of the roof replacement and roof repair options outlined herein are dependent on Gale's findings regarding the PV array feasibility. This submission is augmented with reduced drawings and a preliminary cost estimate. The preliminary cost estimate is intended to provide APRSD with a rough order of magnitude for the work that has been suggested based on the preliminary observations.

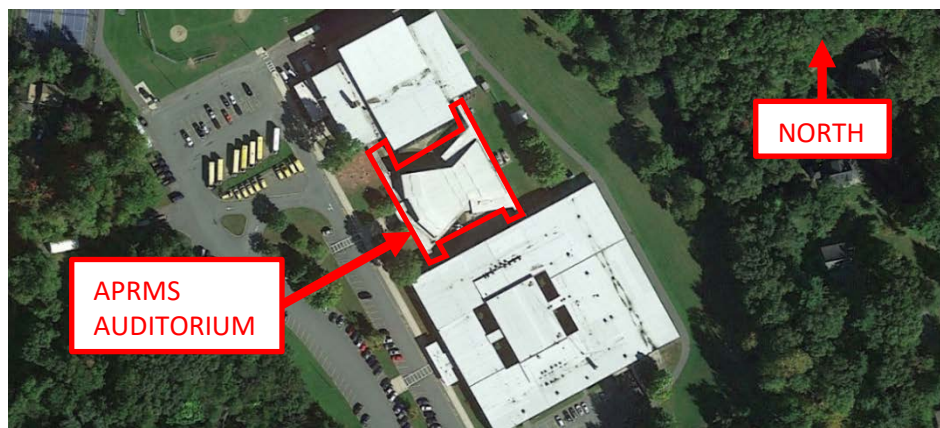


Figure 1: Aerial view of APRMS (image from Google Earth).

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BACKGROUND INFORMATION

APRMS was constructed in the late 1960s and consists of three (3) buildings connected by corridor space. The auditorium is the center building and is surrounded by lower roofs/corridor areas. 1994 and 2001 roof replacement drawings exist for the auditorium roof and adjacent lower roof areas, however, APRMS staff reported to Gale that they do not fully know the extent of the reroofing work that was performed. On the auditorium building, the exterior brick masonry façade runs in a running bond and appear to be standard units (3-5/8" x 2-1/4" x 8"). The current roof system on all five (5) roof areas includes a thermoplastic roof membrane, totaling approximately 17,250 square feet. Heights of these roof areas vary between 10-feet to 20-feet above grade. Note that grade generally appears to be level, however it slopes up to the tree line on the east side of the building.

DOCUMENT REVIEW

Documents provided for Gale's review include the following:

- Regional Junior High School – Amherst-Pelham Regional School District, dated August 14, 1967, by Alderman & MacNeish Architects and Engineers
 - These drawings include foundation and floor plans, elevations, building sections, details, and structural framing plans. These are original construction drawings.
- Roof Replacement Amherst-Pelham Regional Junior High School, dated May 17, 1994, by Tessier Associates, Inc.
 - These drawings include roof area plans, details, and roof framing plans and sections. These drawings include reroofing scope of work.
- Renovations to Amherst-Pelham Regional Middle School, dated April 16, 2001, by Tessier Associates, Inc.
 - These drawings include first floor and roof area plans, foundation and roof details, and plumbing plans. These drawings include reroofing scope of work.

OBSERVATIONS OF EXISTING CONDITIONS

On June 7, 2022, representatives from Gale visited APRMS to conduct a roof evaluation, including interior and exterior observations. Gale, with the assistance of Titan Roofing, Inc. (Titan), also made test cuts in the low-slope and steep-slope thermoplastic roofs to document existing construction. During our evaluation of the existing roof system, Atlas Technical Consultants LLC (Atlas) took samples of various roofing and flashing materials suspected to be asbestos containing materials (ACM).

Observations from these site visits are summarized as follows:

INTERIOR OBSERVATIONS

Gale's interior leak audit consists of information gathered from interviewing building maintenance personnel to document known water infiltration issues and additional observations made by Gale to document other potential water infiltration locations. This information is further identified in Appendices A and B, which include floor and roof plans identifying the reported leak (RL) and observed water stain (WS) locations. Representative photos and more detailed observations are provided herein.

Leak Audit and Observations

Based on discussions with APRMS facilities staff, it is Gale's understanding that the building has experienced moisture infiltration at isolated locations in the auditorium. Past and current reported leak locations exist below Roof Area A (the auditorium roof). APRMS noted that there is one (1) current active leak that occurs every time it rains. APRMS also noted that there were four (4) other recent leak locations that are not as active which may have been addressed by maintenance repairs. Miscellaneous water-stains were also observed in the auditorium and corridors, however, APRMS indicated that these are not known to be active leaks. The reported leak and water stain locations have been projected onto a roof plan (Appendix B), many of which appear to be in the vicinity below the snowguards (refer to the Exterior Observations Section for additional information).



Figure 2: View of the auditorium below Roof Area A.



Figure 3: View of the stained/damaged ceiling at Leak Location #1 in the auditorium.



Figure 4: View of the stained/damaged ceiling at Leak Location #5 in the auditorium.



Figure 5: View of the stained ceiling tiles at Water Stain #3; APRMS indicated that this is not known to be an active leak.

EXTERIOR OBSERVATIONS

On June 7, 2022, representatives from Gale performed a roof evaluation consisting of up-close general observations and limited destructive test cut observations of the existing roof system components on the low-sloped corridor roofs and steeper-sloped auditorium roof. Labels for each roof area are provided in the below figure and in Appendix B; Gale will use these roof area labels when describing exterior observations.

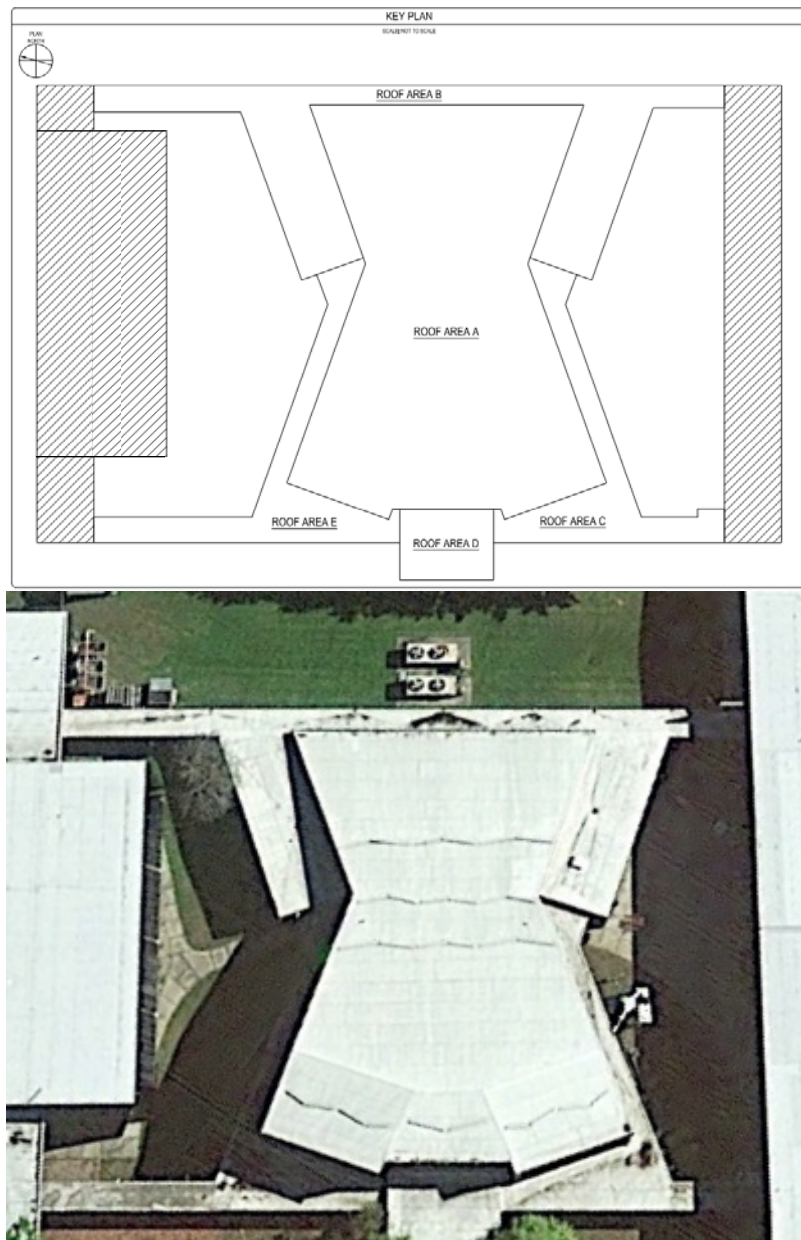


Figure 6: Building Key Plan (aerial image from Google Earth)

Roof General Observations

1. A thermoplastic roof system (totaling approximately 17,250 square feet) exists on five (5) roof areas (Roof Areas A through E). The 60-mil thermoplastic membrane is approximately twenty (20) years old and appears to be in fair condition given its age.



Figure 7: Partial view of Roof Area A.

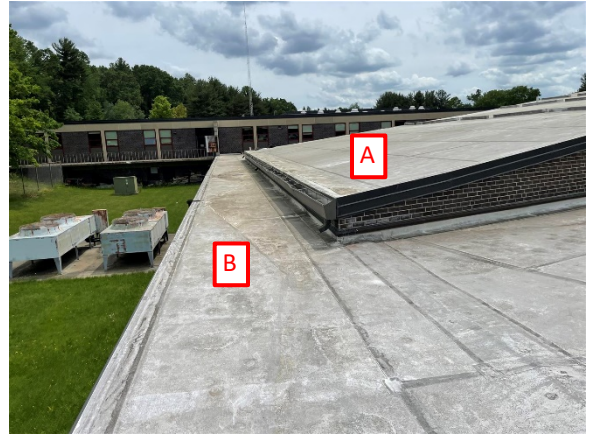


Figure 8: Partial view of Roof Areas A and B.



Figure 9: Partial view of Roof Area C.



Figure 10: Partial view of Roof Area E.

2. There are several wood blocking snow guards that span across Roof Area A in four (4) rows. The wood blocking is secured to the steel framing below and is wrapped in the thermoplastic flashing membrane. The upslope flashing membrane at the snow guards is reversed lapped. The construction of the snow guards is described in more detail in the Roof Destructive Test Cut Observations Section.



Figure 11: Partial view of the snowguards on Roof Area A; note the flashing reverse lap upslope of the snowguards.



Figure 12: Partial view of the snowguards on Roof Area A; note the flashing reverse lap upslope of the snowguards.

3. Roof Area A drains into gutters/downspouts that discharge onto the lower roofs. Roof Areas C and E drain into scuppers/downspouts that discharge onto grade. Roof Areas B and D have internal cast iron roof drains however, there are scuppers/downspouts on the east end of Roof Area B that discharge onto grade. Gale did not evaluate the plumbing for the roof drains.



Figure 13: View of a typical roof drain.



Figure 14: View of a typical scupper and downspout.



Figure 15: Partial view of the gutter on Roof Area A.



Figure 16: View of a broken concrete splash block at the base of an isolated downspout.

4. There are throughwall and reglet flashings at the roof to brick masonry rising wall transitions that vary from four to eleven inches (4"-11") above the existing roof system. Mortar weeps at the throughwall flashing vary between three to six feet (3'-6') on center. Some wall sections do not appear to have weeps. The throughwall flashings appear to be similarly constructed as shown in the original construction drawings. It appears that newer counterflashing has been utilized to clip onto the apparently original throughwall flashing. The sealant at the reglet flashings is debonded. The limits of the throughwall flashing typically stops flush with the roof edge and does not contain a reliable termination. Gale observed locations where the membrane was peeling away from the brick masonry wall.



Figure 17: View of the limits of the throughwall flashing; note that the throughwall flashing stops at the roof edge and the membrane is unadhered in isolated section.



Figure 18: View of the limits of the throughwall flashing; note that the throughwall flashing stops at the roof edge and the membrane is unadhered in isolated section.



Figure 19: View of the throughwall flashing at the precast concrete rising wall from Roof Area E to Roof Area D.



Figure 20: View of a location where the throughwall flashing steps up; note that the upper throughwall flashing is sealed and there does not appear to be weeps or end dams.

5. Brick masonry rising walls exist between the various roof areas. The brick masonry in these locations generally appear to be in good condition with isolated areas of deteriorated mortar and step cracked masonry. The masonry control joints generally appear to be filled with mortar instead of a flexible material (e.g., sealant); the mortar at these locations is typically deteriorated.



Figure 21: View of step cracked masonry between Roof Area B and Roof Area C.



Figure 22: View of deteriorated/missing mortar at a rising wall control joint.



Figure 23: View of deteriorated/missing mortar at a rising wall control joint.



Figure 24: View of a deteriorated mortar joint.

6. Gale was provided access to the roof through the Roof Area B hatch. The roof access ladder is fastened into the CMU wall and into the wood blocking. Gale observed a missing fastener at the upper bracket.



Figure 25: View of the roof access hatch on Roof Area B.



Figure 26: View of the interior roof access ladder; note that there was a missing fastener at the upper bracket.

7. Gale observed isolated brick piers below the roof edge of Roof Areas C and E that were covered with sheet metal that tucked under the roof edge fascia.



Figure 27: View of the masonry piers below Roof Area C.



Figure 28: View of the masonry piers below Roof Area E.



Roof Destructive Test Cut Observations

Gale observed roof destructive test cuts at twenty (20) locations to identify and evaluate the condition and as-built configurations of the representative roof system installations. The test cuts were performed/patched by Titan. Atlas retained some samples at the test cut locations for hazardous material testing; refer to the Hazardous Material Testing Section for additional information. Please be reminded that these test cuts represent the typical construction details but may not be all-inclusive of the construction configurations elsewhere on the building roofs. Refer to Appendix B for the approximate test cut locations. The following conditions summarize our findings:

1. Test Cuts 1 through 20 typically consist of the following construction (from bottom-to-top):
 - o Metal roof deck
 - Roof Areas A, D and portions of B have 1-1/2" thick metal roof decks
 - Roof Areas B, C, and E have 4-1/2" thick metal roof decks
 - Roof Area A has an existing structural framing slope of approximately 1-1/2" per foot.
 - Roof Area B has an existing structural framing slope that varies between 0 to 1/8" per foot.
 - Roof Areas C, D, and E do not appear to have existing structural framing slope and slope is achieved with tapered insulation.
 - o At three (3) locations, Gale observed a 2" thick lightweight concrete fill and asphaltic coating adjacent to the gutter line. This existed on Roof Area B where it abutted the east rising wall of Roof Area A, Roof Area A along the east roof edge, and Roof Area A along the west roof edge.
 - o Loose-laid polyethylene vapor retarder sheet. Gale did not observe a vapor retarder at some Roof Area B opening locations. Where the vapor retarder was present, it appeared to stop a couple of inches short of the roof edge, rising wall, or transition detail.
 - o Mechanically fastened 3" to 7" thick polyisocyanurate rigid insulation. Tapered insulation was observed on Roof Areas A, B, C, and D.
 - o Adhered 60 mil thermoplastic membrane. Gale did not observe any wet materials in the roof test cuts.



Figure 29: View of Test Cut #10 in the field of Roof Area D.



Figure 30: View of Test Cut #2 in the field of Roof Area B; note that there was no vapor retarder observed.

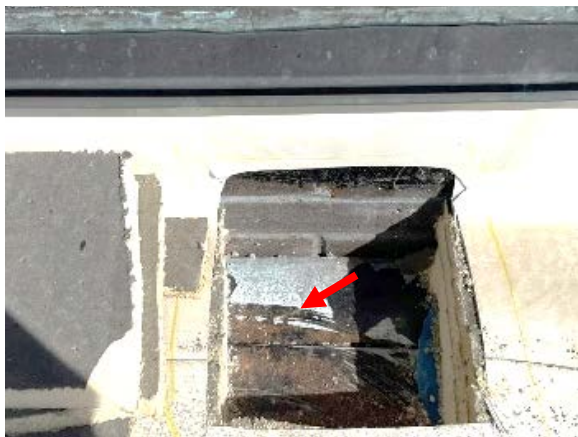


Figure 31: View of a typical rising wall detail; note the vapor retarder stopping short of the rising wall.



Figure 32: View of a typical roof edge detail; note the vapor retarder stopping short of the wood blocking.



Figure 33: View of Test Cut #18 at a snowguard on Roof Area A.

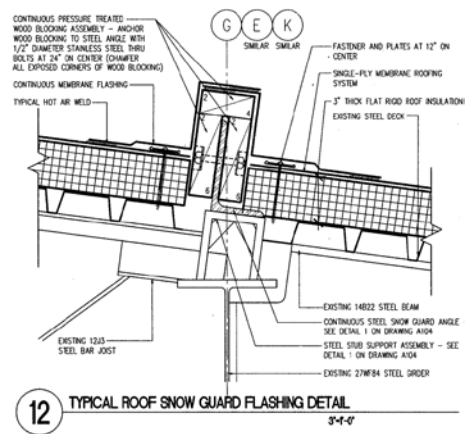


Figure 34: Snow guard detail shown in the "Roof Replacement Amherst-Pelham Regional Junior High School" by Tessier Associates, Inc., dated May 17, 1994.



Figure 35: View of Test Cut #12 at the transition between Roof Area B to Roof Area A.



Figure 36: View of the 2" lightweight concrete fill on Roof Area A.



Figure 37: View of Test Cut #17 at the south roof edge of Roof Area A; note the vapor retarder stopping short of the wood blocking.



Figure 38: View of the test cut between Roof Area C and D.



HAZARDOUS MATERIAL TESTING

Gale retained Atlas to perform sampling of the existing roof components in an attempt to identify if asbestos containing materials (ACM) would be encountered as part of the proposed roof replacement or repair work. Refer to Appendix C for the Hazardous Materials Report, dated June 29, 2022, for more detailed information.

Based on the limited sampling that was performed, no asbestos containing materials were discovered. It should be noted that interior testing was not included within the scope of this evaluation. However, Atlas's report notes the following regarding ACM and the auditorium ceiling:

"According to previous test results within the school's AHERA Management Plan, asbestos-containing textured ceiling material is present above the seating areas of the Auditorium. The material is considered 'friable' which could become dislodged or airborne as a result of vibrations from removal and installation of the new roofing. Additionally, the school has performed emergency spot-abatement projects to fix water damaged areas in the ceiling due to water infiltration from the existing roof."

Additional testing may be required depending on the final design scope of work should disruption of interior materials be required. Should any other ACM be discovered in the future, ACM will need to be contained and removed prior to the removal of the surrounding materials by a licensed Massachusetts Asbestos Contractor. Industrial hygienists typically recommend that an asbestos-removal design be developed during the design phase. This often includes specification section(s) and drawings showing the approximate limits of the materials needing to be abated.

ROOF DISCUSSION AND DESIGN CONSIDERATIONS

It is Gale's opinion based on our evaluation observations, the age of the roof systems, and the reported water leakage occurrences, that the five (5) thermoplastic roof areas are still in a serviceable condition but should be replaced in the next 3 to 5 years. If APRMS is not installing PV arrays, repairs should be considered in an attempt to stop the reported active water leakage and to further extend the service life of the roof. However, if APRMS is going to install new PV arrays on the roof, the roof system should be replaced as it will be more difficult and costly to replace the roof with the PV arrays in place. Since the age and general condition of all five (5) roof areas appear to generally be similar, APRMS should consider replacing the roof system at each roof area, including those that do not have PV arrays installed. However, if budget becomes an issue, Roof Area A should be considered priority since water leakage has been reported beneath this roof only.

The preferred solutions for the roof replacement and repair options are described below. Refer to the preliminary cost estimate section of this report for the anticipated cost and specific scope of each option.



ROOF REPLACEMENT OPTION

Remove the existing adhered thermoplastic roof membrane, insulation, loose-laid vapor retarder (where present), and associated components down to the existing roof deck, and install an adhered thermoplastic roof membrane, adhered high density coverboard, mechanically attached polyisocyanurate insulation, vapor retarder, and baseboard. The thickness of the proposed new roof assembly is anticipated to increase approximately 3-inches to meet the minimum R-Value (R-30) requirements set by the 9th Edition of the Massachusetts State Building Code (MSBC).

1. Roof Membrane Options: Gale has reviewed several options for the roof membrane, including but not limited to, Gravel Surfaced Built-Up Roofing, SBS membrane, PVC/Elvaloy membrane, EPDM membrane, and TPO membrane systems. A brief description for each of these membrane systems is described below:
 - A. Single-ply Thermoplastic (Polyvinyl Chloride – PVC, or Elvaloy based): PVC/Elvaloy membrane systems are available from several manufacturers. Based on our experience, it is Gale's opinion that the reinforced coated systems offer satisfactory chemical/physical properties. PVC/Elvaloy membranes can be installed as a fully adhered, mechanically attached or a loose laid and ballasted application. The PVC/Elvaloy membrane is manufactured in wide rolls. However, the PVC/Elvaloy membrane seams are thermally fused by hot air welds only to form a monolithic sheet that does not rely on adhesives for a watertight bond. Manufacturers of this type of system typically offer ten (10) to twenty (20) year materials and workmanship warranties and have recently promoted thicker membranes that carry a thirty (30) year warranty (including fleece-backed PVC/Elvaloy which also have improved puncture resistance). PVC/Elvaloy warranties do not typically contain a ponded water exclusion as a result of the thermally fused seams. The puncture resistance of the PVC/Elvaloy membrane is lower than built-up roof systems, but they are also easily repairable by using handheld heat welding equipment. Gale is aware of several membrane assemblies in New England which have been performing successfully for over twenty (20) years. There are European installations of these products reported to be in excess of twenty-five (25) years old. Additional walkway pads should be considered on this membrane as it becomes very slippery when wet. Also, Sarnafil has photovoltaic tray brackets that can be heat welded to the thermoplastic membrane that can be considered during the design phase.



B. Gravel Surfaced Built-Up Roofing (GSBUR): This membrane type includes alternating layers of asphalt, reinforcing felts (organic or non-organic) and gravel surfacing. Of the systems discussed herein for low slope applications, the GSBUR is the most time proven. Properly designed and installed, these systems have shown good longevity. GSBUR systems are field fabricated and therefore, considered more workmanship dependent, and can be susceptible to problems during construction. GSBUR can be applied with hot asphalt or cold mastics. Hot applied systems are accompanied by the odor of asphalt and use of 450°F to 500°F asphalt on the roof. The asphalt acts as the waterproofing materials while the fabric and felts provide the strength. Hot asphalt used to install the system can have logistic implications due to the strong fumes associated with the asphalt and the possibility of a fire hazard created by the asphalt kettle application. "Cold-process" built-up systems avoid temperature application of asphalt, and as such, have less odor. The cold process systems offer superior resistance to vandalism but can be difficult to repair. The introduction of the R-30 insulation system under this roof configuration could result in crazing of the asphalt and felt plies during extreme cold weather conditions. Manufacturers of this type of system offer ten to twenty (10-20) and sometimes thirty (30)-year material and workmanship guarantees.

2. Other Design/Scope Considerations:

- A. Wind Uplift: Wind uplift for low-slope roof areas should be performed in conformance with both the MSBC, which references the American Society of Civil Engineers (ASCE) design standard ASCE 7-10, and the Factory Mutual (FM) Global Method, as noted within Data Sheet 1-28, "Wind Design." Increased attachment methods are typically provided at the perimeter and corners of each roof area replaced to withstand the increased uplift pressures for these specific areas. It is Gale's understanding that the facility is not insured by FM so design conformance with FM is not required, unless the MSBC prevails. Final design parameters are recommended to comply with FM, as it is more conservative than the MSBC for this application and is considered an industry standard by roof design professionals.
- B. Roof Pitch and Roof Edge/Rising Wall Conditions: The International Building Code (IBC) outlines requirements where reroofing is required, including drainage of low-sloped roofs, structural and construction activity load requirements, and flashing requirements. Although reroofing shall not be required to meet 1/4 inch per foot slope, positive slope to drains must be provided, or a ponding instability structural analysis is required to confirm that the roof structure possesses adequate stiffness to preclude progressive deflection in the event that the primary drain system becomes obstructed. Roof surfaces which have at least 1/4 inch per foot slope are not considered to be susceptible to ponding instability. Isolated roof areas appear to have a slight slope, less than 1/8 inch per foot. Therefore, it is Gale's opinion that the future design intent includes providing 1/4 inch per foot slope using



tapered insulation throughout Roof Areas B through E. As the overall thickness of the insulation and roof assembly are increased, roof edge and rising wall conditions will require modification. Where roof drains are present, the height of the perimeter wood blocking and sheet metal edge fascia will also need to be increased. Additional wood blocking will need to be installed to match the maximum height of insulation and installing new sheet metal edge fascia to terminate the perimeter of the roof. Note that the new sheet metal fascia may incorporate a 2 or 3-piece fascia system, which will change the aesthetics of the existing edge profile. The flashing at the masonry piers below Roof Areas C and E should also be removed and replaced. Where roof edge gutters and scuppers exist, rising wall conditions will be impacted by the tapered insulation. This typically requires raising the throughwall or reglet flashing height to achieve the industry standard 8 inch minimum height above the roof finished surface, or potentially overcladding the rising wall if the remaining exposed existing rising wall cladding is minimal.

- C. Roof Access: Remove and replace the existing roof access hatch and interior access ladder located on Roof Area B.
- D. Masonry Rising Wall Control Joints: The brick masonry vertical control joints located on Roof Areas B, C, and E have missing or failed mortar joints. It is Gale's opinion that the mortar in these control joints is replaced with an elastomeric sealant under the scope of a roof replacement project to improve the movement capability of the brick masonry and to improve moisture infiltration resistance of the wall assembly. Louvers also exist on some of the rising walls; perimeter sealant at louvers should be removed and replaced.
- E. Roof Drains: Remove existing roof drains and install new bowls with extendable collars, clamps, and cast-iron strainers.
- F. Gutters, Scuppers, and Downspouts: Remove and replace existing gutters, scuppers, and downspouts. Provide roof crickets between these drainage components to promote drainage and manage water.
- G. Fall Protection: As there is no rooftop equipment in the vicinity of the roof edge, portable rails should be considered only at the perimeter of the roof access hatch on Roof Area B. However, should PV arrays be considered, additional fall protection will be required to allow for routine maintenance access; further review of the PV array layout will be required.
- H. Vent Pipes: Vent pipes throughout the roof area are typically less than eighteen inches (18") in height. Given the increased thickness of the new roof system, all vent pipes should be raised to provide eighteen to twenty-four inches (18" to 24") clearance above the finished surface of the new roof system.



- I. Existing Steel Deck: It is Gale's opinion that any sections of the structural steel deck, which appear to be loose or poorly connected to the structural framing, be re-secured as part of this roof replacement project. Since the potential extent of unsecured decking is unknown at this time, it is Gale's opinion that the perimeters and corners of the roof be re-secured as part of the Contract Documents, and any missing fasteners/welds in the field of the roof will likely be covered by a unit cost, should this unforeseen condition be discovered during construction.
 - J. Snow Guards: Modify snow guards as required to accommodate the increased thickness of the new roof system.
3. New Roof System Warranty: Some roof membrane manufacturers, like Sarnafil, have additional warranty stipulations if PV arrays are installed on a newer roof. Depending on the roof membrane manufacturer, some stipulations may include additional manufacturer inspections, specific roof membrane product enhancements (e.g., thicker membrane), installation of an approved coverboard, having the PV arrays installed within five (5) years of the new roof installation, etc. These warranty stipulations should be determined at the beginning stages of a roof replacement design.

ROOF REPAIR OPTION

Perform repairs on the existing roof, rising walls, etc., as noted herein.

- 1. Addressing Reported Water Leakage: The reported water leakage source is unknown at this time. On low-and-moderately sloped roofs, the leak source may exist directly above the interior leak/staining, however, roof leak sources are often located some distance away. For example, defects in low slope roofs that allow water into the roof assembly may result in the water travelling a fair distance along the coverboard, insulation, and deck materials before finding the path to the interior. The same is true for steep slope roofs, however, the leakage source is usually more directly upslope of the interior leak/stain whereas leakage source for low-slope roofs can come from any direction.

Water testing studies must be performed to identify the leak sources so that the appropriate repairs can be applied. This can often result in numerous repair attempts if multiple water leakage sources exist for a given interior leak/water stain location. Any repair should be water tested to confirm the effectiveness of the repair, and the interior area should continue to be monitored during future rain events. Gale can perform water testing and design repairs, if requested.



Since Gale has not performed any water testing at this time, Gale can only provide assumptions on repairs that may or may not address the reported water leakage. By superimposing the APRMS-reported water leakage locations on the roof plan, the interior leaks generally appear to align with the snowguards. Per discussions with APRMS, significant snowdrifts occur at the snow guards. Since the flashing at these locations contain an upslope reverse lap, it is possible that the leading edge of the flashing contains defects, allowing water from the snowdrifts to leak into the roof system. It is Gale's opinion that adding additional flashing/protection measures at these snowguard locations is a good start to a trial and error repair approach.

2. Other Repair Considerations:

- A. Masonry Rising Wall Control Joints : As discussed in the Roof Replacement Option above, the brick masonry vertical control joints located on Roof Areas B, C, and E appear to have missing or have failed mortar joints. It is Gale's opinion that the mortar in these control joints is replaced with an elastomeric sealant under the scope of a repair project to improve the movement capability of the brick masonry and to improve moisture infiltration resistance of the wall assembly. Louvers also exist on some of the rising walls; perimeter sealant at louvers should be removed and replaced.
- B. Masonry Repairs: Rebuild step-cracked brick masonry observed on the rising wall transitions between Roof Areas C/E and B.
- C. Throughwall/Reglet Flashing Terminations: Terminate roof membrane and/or sheet metal with a blind nailer beyond the roof edge and at rising wall inside corners to reduce the potential for water infiltration on this vertical edge.
- D. Splash Block: Remove and replace the deteriorated splash block on Roof Area E, place new splash block on a walkway pad or protection layer.
- E. Roof Access: Remove and replace the existing interior access ladder located at the Roof Area B roof hatch.

PRELIMINARY COST ESTIMATE

The cost estimates presented below are compiled from various sources and are based on our experience with projects similar to this type. The estimates are preliminary as the actual construction schedule, scope of work and construction details have not been fully defined. These estimates are intended to give the APRMS an indication as to what construction costs may be given the existing conditions and Gale's opinions.



Please be reminded that construction costs are very volatile at the time of this report as a result of the recent COVID-19 pandemic, labor shortages, supply delays, and the Ukraine/Russia current conditions. The actual bid results may vary depending on the construction schedule and world events at the time the bid documents are available.

Gale has broken down these estimates based on the repairs and replacement options presented for consideration.

The unit costs presented below typically take into account the Contractor's costs such as:

- Mobilization
- Access
- Crane usage
- Dumpsters
- General oversight of the project
- Bonds and Insurance
- Overhead and Profit

We have also included a twenty percent (20%) contingency within the preliminary construction budget cost in the event that unforeseen conditions are encountered during the design development phase or during construction.

Description of Work	Recommended Budget	Recommended Budget with 20% Contingency
Roof Replacement Option for Roof Areas A-E ¹	\$ 828,000	\$ 993,600
Roof Repair Option	\$ 23,500	\$ 28,200

¹The preliminary cost estimate for the roof replacement option does not include costs associated with the installation of PV arrays or any necessary structural strengthening to the roof framing system. Refer to Gale's forthcoming structural evaluation letter for additional information.

These estimates do not include costs for hazardous material abatement at this time. Should the friable asbestos-containing textured ceiling material be disturbed above the seating areas of the Auditorium, additional abatement costs will need to be included.

Please be reminded that these budgets are preliminary and should not be used for sensitive funding. Also note that these values do not include soft costs associated with working for the APRSD, design/engineering costs, interior repairs/renovations, etc. More refined cost estimates can be provided should the APRSD wish to proceed to the next design phase.

Mr. Rupert Roy-Clark
Amherst-Pelham Regional Middle School Roof Evaluation
Amherst, MA
July 1, 2022
Page 20



Thank you for your attention. Should you have any questions or comments, please do not hesitate to contact us at this office.

Best regards,

GALE ASSOCIATES, INC.

Katie Ciaramitaro/pmww

Katie Ciaramitaro, CDT
Senior Staff Engineer

John W. Kurpeski/pmww

John W. Kurpeski, P.E.
Project Manager

KRC/JWK:pmw

Attachments

cc: Gale Team (CM, JWK, KRC, HNL)

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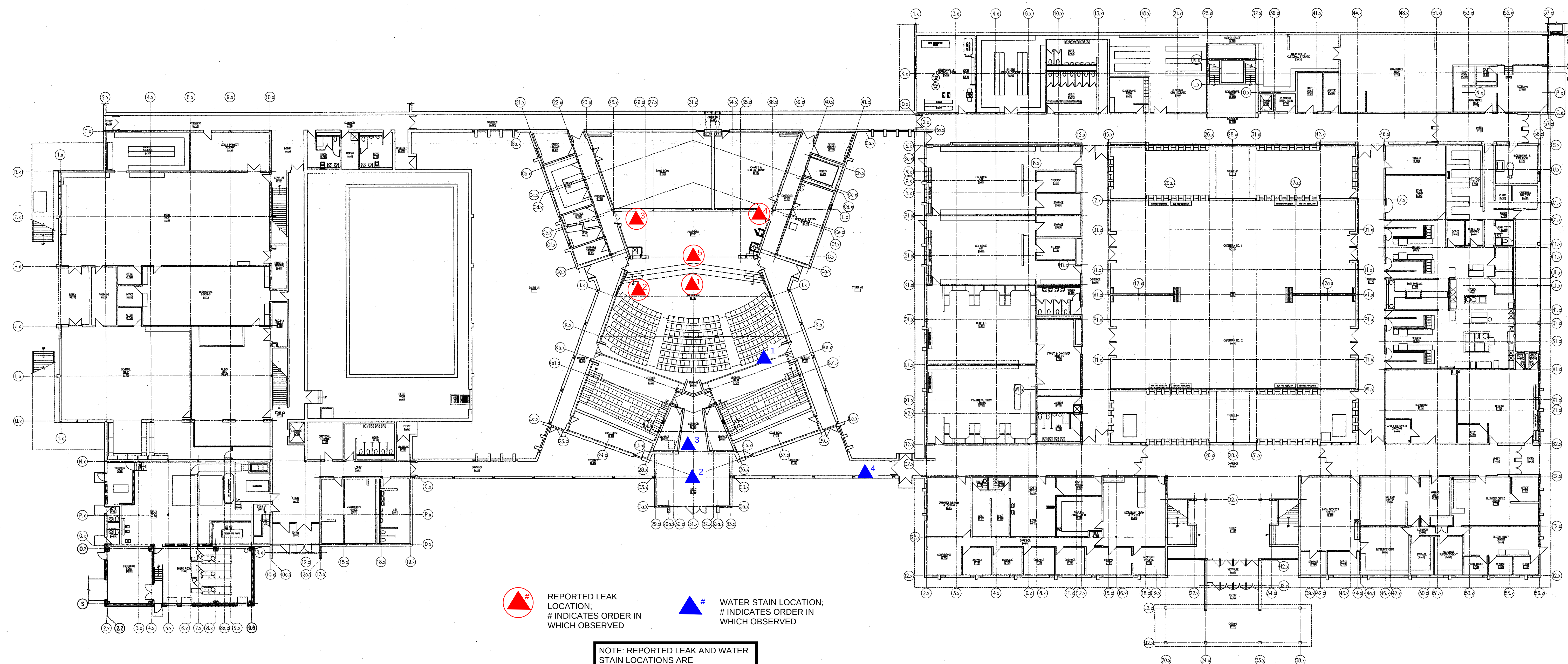


APPENDIX A

INTERIOR LEAK AUDIT OBSERVATIONS

APPENDIX A - INTERIOR LEAK AUDIT OBSERVATIONS

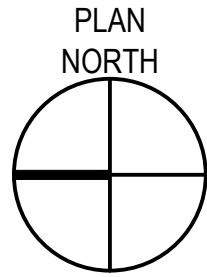
GALE JN: 839810
SHEET 1 OF 1



DRAWING NOTE: FLOOR PLANS
OBTAINED FROM RENOVATIONS TO
AMHERST-PELHAM REGIONAL MIDDLE
SCHOOL, FROM TESSIER ASSOCIATED,
INC., DATED APRIL 16, 2001

APPENDIX B

REDUCED DRAWINGS



APPENDIX B - REDUCED DRAWINGS

GALE JN: 839810
SHEET 1 OF 1



Gale Associates, Inc.
Engineers and Planners
163 LIBBEY PARKWAY
WEYMOUTH, MA 02189
P 781.335.6465 F 781.335.6467
www.gainc.com
Boston Baltimore Orlando Hartford
Manchester

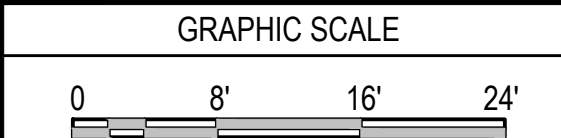
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EVALUATION

PROJECT
APRMS AUDITORIUM ROOF EVALUATION
170 CHESNUT STREET, AMHERST
MASSACHUSETTS 01002

OWNER
AMHERST-PELHAM REGIONAL HIGH SCHOOL
170 CHESNUT STREET, AMHERST
MASSACHUSETTS 01002

NO.	DATE	DESCRIPTION	BY
PROJECT NO.	839810		
CADD FILE	839810 A100s		
DESIGNED BY	JWK		
DRAWN BY	HNL/KRC		
CHECKED BY	CM		
DATE	07/01/2022		
DRAWING SCALE	3/32"=1'-0"		



SHEET TITLE

ROOF AREA
PLAN

DRAWING NO.

A101

SHEET NOTES

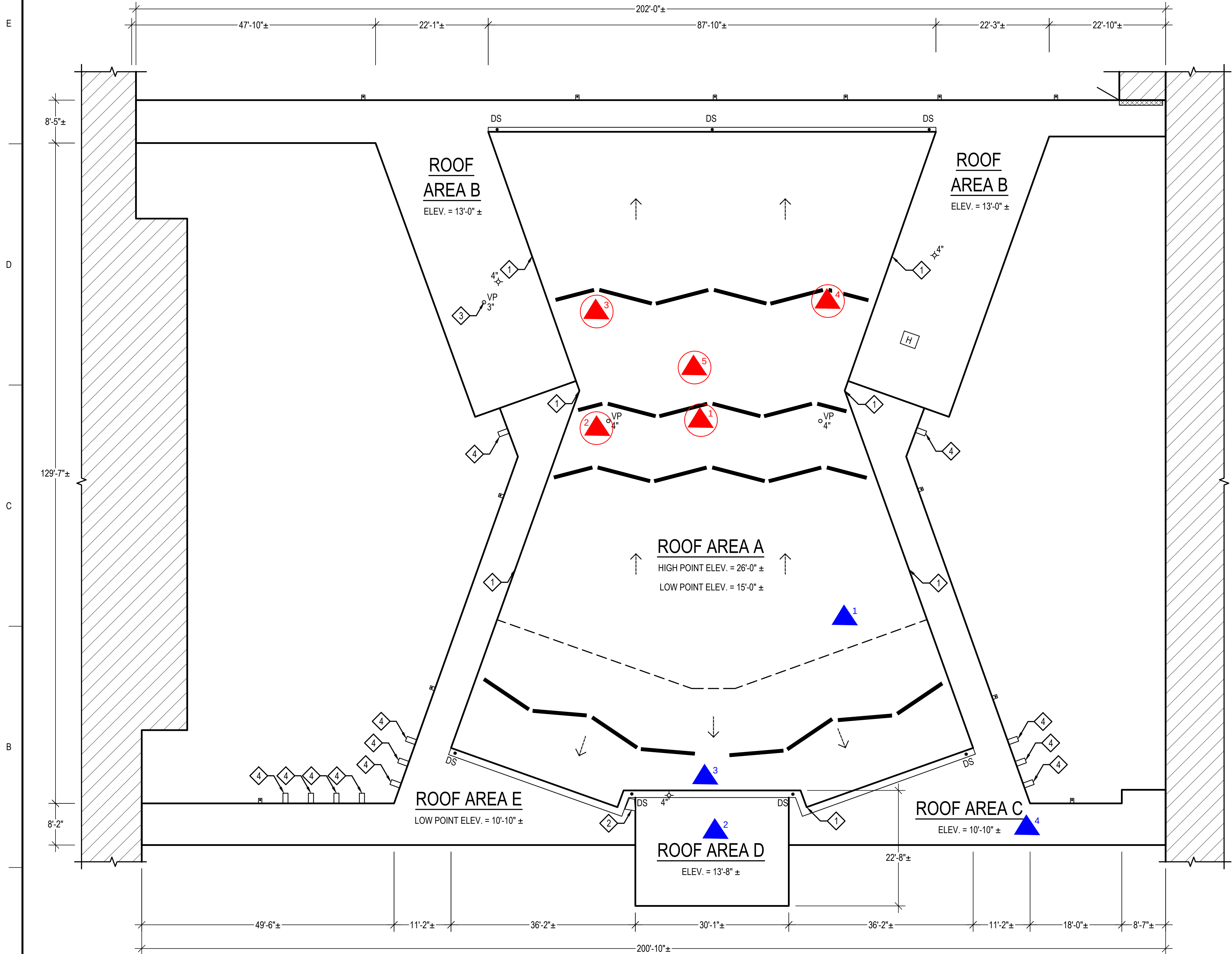
- 1 FAILED RISING WALL EXPANSION JOINT; APPROXIMATELY 15 LINEAR FEET
- 2 DETERIORATED SPLASH BLOCK
- 3 CLOGGED VENT PIPE
- 4 BRICK MASONRY COLUMNS BELOW ROOF LINE

LEGEND

- ALL ITEMS ARE EXISTING UNLESS OTHERWISE INDICATED.
- ROOF EDGE
 - - - RIDGE/HIP LINE
 - DS DS GUTTER WITH DOWNSPOUT
 - SNOW GUARD
 - [Hatched] BUILDING NOT IN CONTRACT (N.I.C.)
 - [Triangle] SPLASH BLOCK
 - [Cross-hatched] CONCRETE BLOCK RISING WALL
 - METAL SCREEN WALL
 - [Red Triangle #] REPORTED LEAK LOCATION SUPERIMPOSED ON ROOF PLAN; # INDICATES ORDER IN WHICH OBSERVED
 - [Symbol] ROOF DRAIN; # INDICATES DIAMETER
 - [Symbol] VENT PIPE; # INDICATES DIAMETER
 - [Arrow] STRUCTURAL SLOPE
 - [Symbol] SCUPPER AND DOWNSPOUT
 - [H] ROOF HATCH
 - [Blue Triangle #] WATER STAIN LOCATION SUPER IMPOSED ON ROOF PLAN; # INDICATES ORDER IN WHICH OBSERVED

GENERAL NOTES

- THE INFORMATION SHOWN ON THIS DRAWING HAS BEEN COMPILED FROM VARIOUS SOURCES AND MAY NOT REFLECT THE ACTUAL CONDITIONS AT THE TIME OF CONSTRUCTION.
- FOR THE SAKE OF CLARITY, EACH INDIVIDUAL DETAIL ON THE ROOF PLAN HAS NOT BEEN INDICATED. INSTALLATION DETAILS HAVE BEEN INDICATED FOR TYPICAL COMPONENTS AT RANDOM LOCATIONS.
- HATCH PATTERNS ARE FOR REPRESENTATION ONLY AND SHOULD NOT BE USED AS A MEANS FOR QUANTIFYING.
- JOB SITE SAFETY IS THE CONTRACTOR'S RESPONSIBILITY. GALE REPRESENTATIVES, INCLUDING SUBCONSULTANTS RETAINED BY GALE, MAY VISIT THE JOB SITE FROM TIME TO TIME. THESE VISITS ARE FOR CLARIFICATIONS OF SPECIFIC DESIGN RELATED ISSUES ONLY AND ARE NOT FOR THE PURPOSES OF JOB SITE SAFETY. IT IS THE CONTRACTOR'S SOLE RESPONSIBILITY TO COMPLY WITH ALL SITE SAFETY APPLICABLE REQUIREMENTS.
- ALL ITEMS OF CONSTRUCTION SHALL BE IN ACCORDANCE WITH LOCAL, STATE AND FEDERAL REGULATIONS APPLICABLE TO THE PROJECT.
- THE CONTRACTOR SHALL REPORT DETERIORATED OR UNSUITABLE STRUCTURAL DECK OR SUBSTRATES TO THE OWNER PRIOR TO PERFORMING ROOFING INSTALLATION WORK.
- THE CONTRACTOR IS CAUTIONED THAT DUE TO BUILDING OCCUPANCY, THE OWNER REQUIRES COMPLIANCE ON WORK HOURS, SCHEDULING, SET-UP, CLEANUP, PARKING, SECURITY, ETC. REFER TO SPECIFICATIONS FOR OWNER REQUIREMENTS.
- NOTIFY THE OWNER OF WORK LOCATIONS A MINIMUM OF 72 HOURS IN ADVANCE IN ORDER THAT OCCUPANCY BELOW THE WORK AREAS MAY BE CONTROLLED.
- ROOFING AND FLASHING SHALL BE 100% WEATHER TIGHT ON A DAILY BASIS. REMOVE ONLY AS MUCH EXISTING MATERIAL AS CAN BE REPLACED ON THE SAME DAY. PHASED CONSTRUCTION IS NOT PERMITTED.
- DETAILS NOT DEPICTED SHALL BE CONSTRUCTED IN A MANNER CONSISTENT WITH THE DETAIL DRAWINGS.
- REFER TO CONTRACT DOCUMENTS FOR SCHEDULE LIMITATIONS OF EXTERIOR WORK.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR REMOVING AND LEGALLY DISPOSING OF ALL MATERIALS LISTED AND/OR INDICATED ON THE DRAWINGS OR NOTED IN THE WRITTEN SPECIFICATIONS - TYPICAL.
- THE BUILDING SHALL BE MAINTAINED IN A DUST FREE CONDITION. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY TEMPORARY PROTECTION TO ENSURE THE INTERIOR OF THE BUILDING REMAINS DUST FREE. NO DEMOLITION CAN PROCEED UNTIL AN APPROVED TEMPORARY PROTECTION SYSTEM IS IN PLACE.
- DUE TO THE CONGESTION AROUND THE PROJECT SITE THE CONTRACTOR MUST PROVIDE DETAILED TRAFFIC AND SCAFFOLDING PLANS TO THE OWNER FOR REVIEW. THE CONTRACTOR WILL ONLY BE ALLOWED THE AREAS SHOWN FOR SETUP AND STORAGE OF MATERIALS.





APPENDIX C

HAZARDOUS MATERIAL TESTING REPORT



June 29, 2022

Mr. John Kurpeski
Project Manager
Gale Associates, Inc.
163 Libbey Parkway
Weymouth, MA 02189

Re: Asbestos Roof Sampling Report
Amherst Middle School
Auditorium Roof Replacement Project
Amherst, Massachusetts

Dear Mr. Kurpeski:

Atlas Technical Consultants, LLC (ATLAS) was retained by Gale Associates, Inc. to perform asbestos roof sampling at the Amherst Middle School located in Amherst, Massachusetts. Specifically, ATLAS's Scope of Work included the following:

1. Performance of a limited Asbestos Inspection for accessible suspect asbestos-containing roofing materials located on the Auditorium section of Amherst High School subject to disturbance by the forthcoming roof removal project.
2. Samples were collected from roof cores that were advanced and repaired by the Roofing Contractor retained by Gale Associates, Inc.
3. All bulk samples were analyzed for asbestos content using Polarized Light Microscopy (PLM) with Dispersion Staining EPA Method 600/R-93-116 per 40 CFR 763. To qualify as asbestos containing, the material must be determined to contain *greater than one percent (>1%)* asbestos from a homogeneous material area set of samples.

The following Table summarizes the results of ATLAS's sampling:

LOCATION	MATERIAL	RESULT
A-1 Lower Roof	1/8" White Membrane	No Asbestos Detected
A-2 Lower Roof	1/8" White Membrane	No Asbestos Detected
B-1 Lower Roof	1/8" White Membrane	No Asbestos Detected
A-1 Lower Roof	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
A-2 Lower Roof	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
B-1 Lower Roof	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
A-1 Lower Roof	ISO Foam Insulation	No Asbestos Detected
A-2 Lower Roof	ISO Foam Insulation	No Asbestos Detected
B-1 Lower Roof	ISO Foam Insulation	No Asbestos Detected
C-Auditorium Roof	ISO Foam Insulation	No Asbestos Detected
A-1 Lower Roof	Residue on Metal Deck	No Asbestos Detected
A-2 Lower Roof	Residue on Metal Deck	No Asbestos Detected
B-1 Lower Roof	Residue on Metal Deck	No Asbestos Detected
C-Auditorium Roof	Residue on Metal Deck	No Asbestos Detected



LOCATION	MATERIAL	RESULT
A-1 Lower Roof Wall Edge	1/8" White Membrane	No Asbestos Detected
A-2 Lower Roof Wall Edge	1/8" White Membrane	No Asbestos Detected
B-1 Lower Roof Wall Edge	1/8" White Membrane	No Asbestos Detected
A-1 Lower Roof Wall Edge	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
A-2 Lower Roof Wall Edge	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
B-1 Lower Roof Wall Edge	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
A-1 Lower Roof Wall Edge	White Felt Under Membrane	No Asbestos Detected
A-2 Lower Roof Wall Edge	White Felt Under Membrane	No Asbestos Detected
B-1 Lower Roof Wall Edge	White Felt Under Membrane	No Asbestos Detected
A-1 Lower Roof Wall Edge	Mastic Under White Felt Membrane	No Asbestos Detected
A-2 Lower Roof Wall Edge	Mastic Under White Felt Membrane	No Asbestos Detected
B-1 Lower Roof Wall Edge	Mastic Under White Felt Membrane	No Asbestos Detected
A-1 Perimeter Wall Edge	1/8" White Membrane	No Asbestos Detected
A-2 Perimeter Wall Edge	1/8" White Membrane	No Asbestos Detected
B-1 Perimeter Wall Edge	1/8" White Membrane	No Asbestos Detected
A-1 Perimeter Wall Edge	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
A-2 Perimeter Wall Edge	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
B-1 Perimeter Wall Edge	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected
A-1 CMU Wall	Clear Caulking on CMU Wall	No Asbestos Detected
A-1 Lower Roof	Yellow Mastic on Roof Clips	No Asbestos Detected
A-2 Lower Roof	Yellow Mastic on Roof Clips	No Asbestos Detected
A-1 Lower Roof Wall Edge	Black Roof Coating	No Asbestos Detected
A-2 Lower Roof Wall Edge	Black Roof Coating	No Asbestos Detected
B-1 Lower Roof Wall Edge	Black Roof Coating	No Asbestos Detected
A-2 Lower East Side	Black Residue on Metal Corrugated Deck	No Asbestos Detected
A-3 Lower Perimeter Edge	Black Coating on Wood Blocking	No Asbestos Detected
A-2 Lower Roof	Black Mop Seal on Concrete Slab	No Asbestos Detected
C-Auditorium – South Side	Black Mop Seal on Concrete Slab	No Asbestos Detected
B-1 Lower Wall Edge	Brown Caulking on Brick Walls	No Asbestos Detected
B-2 Lower Wall Edge	Brown Caulking on Brick Walls	No Asbestos Detected
B-3 – Inner Wall Edge	White Caulking on Brick Walls	No Asbestos Detected
C-Auditorium Roof	Black Seam on Metal Flashing	No Asbestos Detected
C-Auditorium Roof	1/8" White Membrane	No Asbestos Detected
C-Auditorium Roof	Yellow Mastic Under 1/8" White Membrane	No Asbestos Detected

Note: According to previous test results within the schools AHERA Management Plan, asbestos-containing textured ceiling material is present above the seating areas of the Auditorium. The material is considered “friable” which could become dislodge or airborne as a result of vibrations from removal and installation of the new roofing. Additionally, the school has performed emergency spot-abatement projects to fix water damaged areas in the ceiling due to water infiltration from the existing roof.



June 29, 2022

Page 3 of 3

Asbestos Roof Sampling Report
Amherst Middle School
Auditorium Roof Replacement Project
Amherst, Massachusetts

If you have any questions regarding this report, please give Derrick Wissman a call directly at (413) 664-6687.

ATLAS TECHNICAL CONSULTANTS, LLC

A handwritten signature in black ink, appearing to read "Derrick Wissman".

Derrick Wissman
Senior Project Manager
Direct Line 413-664-6687
derrick.wissman@oneatlas.com

A handwritten signature in black ink, appearing to read "Brian Williams".

Brian Williams
Area Manager
Direct Line 413-504-1653
brian.williams@oneatlas.com



June 29, 2022

ATTACHMENT A

Asbestos Roof Sampling Report
Amherst Middle School
Auditorium Roof Replacement Project
Amherst, Massachusetts

ATTACHMENT A

PLM BULK ASBESTOS LABORATORY REPORT



EMSL Analytical, Inc.

307 West 38th Street New York, NY 10018

Tel/Fax: (212) 290-0051 / (212) 290-0058

<http://www.EMSL.com> / manhattanlab@emsl.com

EMSL Order: 032209137

Customer ID: ATC62

Customer PO: 11-81-0030

Project ID:

Attention: Derrick Wissman
Atlas Technical
73 William Franks Drive
West Springfield, MA 01089

Phone: (413) 781-0070
Fax: (413) 781-3734
Received Date: 06/08/2022 10:01 AM
Analysis Date: 06/17/2022 - 06/20/2022
Collected Date: 06/07/2022

Project: Amherst Pelham Regional Middle School / Gale Associates / 170 Chestnut St. Amherst, Ma. / 183DW22062

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-01a 032209137-0001	A-1 Lower Roof - 1/8" White Membrane	Gray/White Non-Fibrous Heterogeneous	3% Glass	97.0% Non-fibrous (Other)	None Detected
AMR-01b 032209137-0002	A-2 Lower Roof - 1/8" White Membrane	Gray/White Non-Fibrous Heterogeneous	3% Glass	97.0% Non-fibrous (Other)	None Detected
AMR-01c 032209137-0003	B-1 Lower Roof - 1/8" White Membrane	Gray/White Non-Fibrous Heterogeneous	3% Glass	97.0% Non-fibrous (Other)	None Detected
AMR-02a 032209137-0004	A-1 Lower Roof - Yellow Mastic under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-02b 032209137-0005	A-2 Lower Roof - Yellow Mastic under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-02c 032209137-0006	B-1 Lower Roof - Yellow Mastic under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-03a 032209137-0007	A-1 Lower Roof - ISO Foam Insulation	Tan Fibrous Heterogeneous	3% Cellulose 2% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-03b 032209137-0008	A-2 Lower Roof - ISO Foam Insulation	Brown/Yellow Fibrous Heterogeneous	4% Cellulose 2% Glass	94.0% Non-fibrous (Other)	None Detected
AMR-03c 032209137-0009	B-1 Lower Roof - ISO Foam Insulation	Brown/Yellow Fibrous Heterogeneous	3% Cellulose 2% Glass	95.0% Non-fibrous (Other)	None Detected

EMSL maintains liability limited to cost of analysis. Interpretation and use of test results are the responsibility of the client. This report relates only to the samples reported above, and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. The report reflects the samples as received. Results are generated from the field sampling data (sampling volumes and areas, locations, etc.) provided by the client on the Chain of Custody. Samples are within quality control criteria and met method specifications unless otherwise noted. The above analyses were performed in general compliance with Appendix E to Subpart E of 40 CFR (previously EPA 600/M4-82-020 "Interim Method") but augmented with procedures outlined in the 1993 ("final") version of the method. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. Non-friable organically bound materials present a problem matrix and therefore EMSL recommends gravimetric reduction prior to analysis. Unless requested by the client, building materials manufactured with multiple layers (i.e. linoleum, wallboard, etc.) are reported as a single sample. Estimation of uncertainty is available on request.

Samples analyzed by EMSL Analytical, Inc. Rochester, NY NVLAP Lab Code 600183-0

Initial report from: 06/21/2022 11:16:36



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307 West 38th Street New York, NY 10018

Tel/Fax: (212) 290-0051 / (212) 290-0058

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Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-03d 032209137-0010	C-Auditorium Roof - ISO Foam Insulation	Brown/Yellow Fibrous Heterogeneous	3% Cellulose 2% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-04a 032209137-0011	A-1 Lower Roof - Residue on Metal Corrugated Deck	Gray Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-04b 032209137-0012	A-2 Lower Roof - Residue on Metal Corrugated Deck				Layer Not Present
AMR-04c 032209137-0013	B-1 Lower Roof - Residue on Metal Corrugated Deck				Layer Not Present
AMR-04d 032209137-0014	C-Auditorium Roof - Residue on Metal Corrugated Deck				Layer Not Present
AMR-05a 032209137-0015	A-1 Lower Roof Wall Edge - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-05b 032209137-0016	A-2 Lower Roof Wall Edge - 1/8" White Membrane	Gray/White/Yellow Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-05c 032209137-0017	B-1 Lower Roof Wall Edge - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-06a 032209137-0018	A-1 Lower Roof Wall Edge - Yellow Mastic Under 1/8" White Membrane	Yellow Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected

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Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-06b 032209137-0019	A-2 Lower Roof Wall Edge - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-06c 032209137-0020	B-1 Lower Roof Wall Edge - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-07a 032209137-0021	A-1 Lower Roof Wall Edge - White Felt Under Membrane	White Fibrous Homogeneous	97% Synthetic	3.0% Non-fibrous (Other)	None Detected
AMR-07b 032209137-0022	A-2 Lower Roof Wall Edge - White Felt Under Membrane	White Fibrous Homogeneous	97% Synthetic	3.0% Non-fibrous (Other)	None Detected
AMR-07c 032209137-0023	B-1 Lower Roof Wall Edge - White Felt Under Membrane	White Fibrous Homogeneous	97% Synthetic	3.0% Non-fibrous (Other)	None Detected
AMR-08a 032209137-0024	A-1 Lower Roof Wall Edge - Mastic White Felt Under Membrane	Yellow Non-Fibrous Heterogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-08b 032209137-0025	A-2 Lower Roof Wall Edge - Mastic White Felt Under Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-08c 032209137-0026	B-1 Lower Roof Wall Edge - Mastic White Felt Under Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-09a 032209137-0027	A-1 Perimeter Wall Edge - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected

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Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-09b 032209137-0028	A-2 Perimeter Wall Edge - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-09c 032209137-0029	B-1 Perimeter Wall Edge - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-10a 032209137-0030	A-1 Perimeter Wall Edge - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-10b 032209137-0031	A-2 Perimeter Wall Edge - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-10c 032209137-0032	B-1 Perimeter Wall Edge - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-11a 032209137-0033	A-1 CMU Wall - Clear Caulk on CMU Wall	White Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-11b 032209137-0034	A-1 CMU Wall - Clear Caulk on CMU Wall	White Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-12a 032209137-0035	A-1 Lower Roof - Yellow Mastic on Roof Clips	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-12b 032209137-0036	A-2 Lower Roof - Yellow Mastic on Roof Clips	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected

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West Springfield, MA 01089

Phone: (413) 781-0070
Fax: (413) 781-3734
Received Date: 06/08/2022 10:01 AM
Analysis Date: 06/17/2022 - 06/20/2022
Collected Date: 06/07/2022

Project: Amherst Pelham Regional Middle School / Gale Associates / 170 Chestnut St. Amherst, Ma. / 183DW22062

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-13a 032209137-0037	A-1 Lower Roof Wall Edge - Black Roof Coating	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-13b 032209137-0038	A-2 Lower Roof Wall Edge - Black Roof Coating	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-13c 032209137-0039	B-1 Lowe Wall Edge - Black Roof Coating	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-14a 032209137-0040	A-2 Lower East Side - Blck Residue on Metal Corrugated Deck	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-14b 032209137-0041	A-2 Lower East Side - Blck Residue on Metal Corrugated Deck	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-15a 032209137-0042	A-3 Lower Permiter Edge - Black Coat on Wood Permimiter Edge	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-15b 032209137-0043	A-3 Lower Permiter Edge - Black Coat on Wood Permimiter Edge	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-16a 032209137-0044	A-2 Lower Roof - Black Mop Seal on Concrete Slab	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-16b 032209137-0045	C-Auditorium on South Side - Black Mop Seal on Concrete Slab	Black Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected

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Samples analyzed by EMSL Analytical, Inc. Rochester, NY NVLAP Lab Code 600183-0

Initial report from: 06/21/2022 11:16:36



EMSL Analytical, Inc.

307 West 38th Street New York, NY 10018

Tel/Fax: (212) 290-0051 / (212) 290-0058

<http://www.EMSL.com> / manhattanlab@emsl.com

EMSL Order: 032209137

Customer ID: ATC62

Customer PO: 11-81-0030

Project ID:

Attention: Derrick Wissman
Atlas Technical
73 William Franks Drive
West Springfield, MA 01089

Phone: (413) 781-0070
Fax: (413) 781-3734
Received Date: 06/08/2022 10:01 AM
Analysis Date: 06/17/2022 - 06/20/2022
Collected Date: 06/07/2022

Project: Amherst Pelham Regional Middle School / Gale Associates / 170 Chestnut St. Amherst, Ma. / 183DW22062

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-17a 032209137-0046	B-1 Lower Wall Edge - Brown Caulk on Brick Walls	Brown Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-17b 032209137-0047	B-2 Lower Wall Edge - Brown Caulk on Brick Walls	Brown Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-18a 032209137-0048	B-3 Inner Wall Edge - White Caulk on Brick Wall	White Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-18b 032209137-0049	B-3 Inner Wall Edge - White Caulk on Brick Wall	White Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-19a 032209137-0050	C-Auditorium Sides - Black Seam on Metal Flashing	Clear Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-19b 032209137-0051	C-Auditorium Roof - Black Seam on Metal Flashing	Clear Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-20a 032209137-0052	C-Auditorium Roof - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-20b 032209137-0053	C-Auditorium Roof - 1/8" White Membrane	Gray/White Fibrous Heterogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-21a 032209137-0054	C-Auditorium Roof - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected

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Samples analyzed by EMSL Analytical, Inc. Rochester, NY NVLAP Lab Code 600183-0

Initial report from: 06/21/2022 11:16:36



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307 West 38th Street New York, NY 10018

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EMSL Order: 032209137

Customer ID: ATC62

Customer PO: 11-81-0030

Project ID:

Attention: Derrick Wissman
Atlas Technical
73 William Franks Drive
West Springfield, MA 01089

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Fax: (413) 781-3734
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Project: Amherst Pelham Regional Middle School / Gale Associates / 170 Chestnut St. Amherst, Ma. / 183DW22062

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
AMR-21b 032209137-0055	C-Auditorium Roof - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-22a 032209137-0056	C-Auditorium Roof - 1/8" White Membrane	Gray/White Non-Fibrous Homogeneous	5% Glass	95.0% Non-fibrous (Other)	None Detected
AMR-22b 032209137-0057	C-Auditorium Roof - 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-23a 032209137-0058	C-Auditorium Roof - Yellow Mastic Under 1/8" White Membrane	Gray/White Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected
AMR-23b 032209137-0059	C-Auditorium Roof - Yellow Mastic Under 1/8" White Membrane	Yellow Non-Fibrous Homogeneous		100.0% Non-fibrous (Other)	None Detected

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Samples analyzed by EMSL Analytical, Inc. Rochester, NY NVLAP Lab Code 600183-0

Initial report from: 06/21/2022 11:16:36



EMSL Analytical, Inc.

307 West 38th Street New York, NY 10018

Tel/Fax: (212) 290-0051 / (212) 290-0058

<http://www.EMSL.com> / manhattanlab@emsl.com

EMSL Order: 032209137

Customer ID: ATC62

Customer PO: 11-81-0030

Project ID:

Attention: Derrick Wissman
Atlas Technical
73 William Franks Drive
West Springfield, MA 01089

Phone: (413) 781-0070

Fax: (413) 781-3734

Received Date: 06/08/2022 10:01 AM

Analysis Date: 06/17/2022 - 06/20/2022

Collected Date: 06/07/2022

Project: Amherst Pelham Regional Middle School / Gale Associates / 170 Chestnut St. Amherst, Ma. / 183DW22062

The samples in this report were submitted to EMSL for analysis by Asbestos Analysis of Bulk materials via EPA/600 (0513) Method using Polarized Light Microscopy. The reference number for these samples is the EMSL Order ID above. Please use this reference number when calling about these samples.

Report Comments:

Sample Receipt Date: 06/08/2022

Sample Receipt Time: 10:01 AM

Analysis Completed Date: 06/20/2022

Analysis Completed Time: 4:50 PM

Analyst(s):

Thomas Hawley PLM (56)

Samples Reviewed and approved by:

Charles Johnson, Asbestos Laboratory Manager
or other approved signatory

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Samples analyzed by EMSL Analytical, Inc. Rochester, NY NVLAP Lab Code 600183-0

Initial report from: 06/21/2022 11:16:36

Relinquished by: <u>[Signature]</u>	Date/Time: 06-07-2022	032209137		
Received by: <u>[Signature]</u>	Date/Time: 6/8/22 10:01 AM			
Relinquished by: <u>[Signature]</u>	Date/Time: _____			
Received by: _____	Date/Time: _____			
Relinquished by: _____	Date/Time: _____			
Company Name: Atlas Technical	EMSL ANALYTICAL, INC 307 West 38th Street New York, NY 10018 (212)-290-0051			
Company Address: 73 William Franks Rd				
City: West Springfield State: MA Zip: 01089				
Phone: (413) 781-0070	Project Name: AmherstPelham Regional Middle School/Gale Associat	Analysis Type: PLM		
Results to: Derrick.wissman@oneatlas.com	Project Address: 170 Chestnut St. Amherst, Ma.	Positive Stop: Yes		
Verbal Results: Yes No	Project Manager: Derrick Wissman	Turnaround Time: 7 Days		
Cell or Pager 1-413-664-6687	Project Number: 183DW22062	Fax Copy by:		
Site Fax:	Sampled by: Jim Lowell Date: 06-07-2022	Return Samples: Yes	No	
Additional Fax:	Special Instructions			
Lab ID	Field ID	Location	Sample Description	Homogenous Area #
	AMR-01a	A-1 Lower Roof	1/8" White Membrane	1
	AMR-01b	A-2 Lower Roof	1/8" White Membrane	1
	AMR-01c	B-1 Lower Roof	1/8" White Membrane	1
	AMR-02a	A-1 Lower Roof	Yellow Mastic Under 1/8" White Membrane	2
	AMR-02b	A-2 Lower Roof	Yellow Mastic Under 1/8" White Membrane	2
	AMR-02c	B-1 Lower Roof	Yellow Mastic Under 1/8" White Membrane	2
	AMR-03a	A-1 Lower Roof	ISO Foam Insulation	3
	AMR-03b	A-2 Lower Roof	ISO Foam Insulation	3
	AMR-03c	B-1 Lower Roof	ISO Foam Insulation	3
	AMR-03d	C-Auditorium Roof	ISO Foam Insulation	3
	AMR-04a	A-1 Lower Roof	Residue on Metal Corrugated Deck	4
	AMR-04b	A-2 Lower Roof	Residue on Metal Corrugated Deck	4
	AMR-04c	B-1 Lower Roof	Residue on Metal Corrugated Deck	4
	AMR-04d	C-Auditorium Roof	Residue on Metal Corrugated Deck	4

Relinquished by: <u>[Signature]</u>	Date/Time: 06-07-2022
Received by: <u>[Signature]</u>	Date/Time: 6/8/22
Relinquished by:	Date/Time:
Received by:	Date/Time:
Relinquished by:	Date/Time:
Received by:	Date/Time:

Company Name: Atlas Technical
 Company Address: 73 William Franks Rd

City: West Springfield State: MA Zip: 01089

Phone: (413) 781-0070

Project Name: AmherstPelham Regional Middle School/Gale Associates
 Project Address: 170 Chestnut St. Amherst, Ma.

Results to: Derrick.wissman@oneatlas.com

Project Manager: Derrick Wissman

Verbal Results: Yes No

Project Number: 183DW22062

Cell or Pager 1-413-664-6687

Sampled by: Jim Lowell

Date: 06-07-2022

Return Samples: Yes

No

Site Fax:

Special Instructions

Additional Fax:

Lab ID	Field ID	Location	Sample Description	Homogenous Area #
	AMR-05a	A-1 Lower Roof Wall Edge	1/8" White Membrane	5
	AMR-05b	A-2 Lower Roof Wall Edge	1/8" White Membrane	5
	AMR-05c	B-1 Lower Roof Wall Edge	1/8" White Membrane	5
	AMR-06a	A-1 Lower Roof Wall Edge	Yellow Mastic Under 1/8" White Membrane	6
	AMR-06b	A-2 Lower Roof Wall Edge	Yellow Mastic Under 1/8" White Membrane	6
	AMR-06c	B-1 Lower Roof Wall Edge	Yellow Mastic Under 1/8" White Membrane	6
	AMR-07a	A-1 Lower Roof Wall Edge	White Felt Under Membrane	7
	AMR-07b	A-2 Lower Roof Wall Edge	White Felt Under Membrane	7
	AMR-07c	B-1 Lower Roof Wall Edge	White Felt Under Membrane	7
	AMR-08a	A-1 Lower Roof Wall Edge	Mastic White Felt Under Membrane	8
	AMR-08b	A-2 Lower Roof Wall Edge	Mastic White Felt Under Membrane	8
	AMR-08c	B-1 Lower Roof Wall Edge	Mastic White Felt Under Membrane	8
	AMR-09a	A-1 Perimeter Wall Edge	1/8" White Membrane	9

EMSL ANALYTICAL, INC

307 West 38th Street
 New York, NY 10018

(212)-290-0051

16.074m

032209137

Relinquished by:	<i>[Signature]</i>	Date/Time:	06-07-2022
Received by:	<i>[Signature]</i>	Date/Time:	6/8/22
Relinquished by:		Date/Time:	
Received by:		Date/Time:	
Relinquished by:		Date/Time:	
Received by:		Date/Time:	

Company Name:	Atlas Technical
Company Address:	73 William Franks Rd
City:	West Springfield State: MA Zip: 01089

EMSL ANALYTICAL, INC
307 West 38th Street
New York, NY 10018
(212)-290-0051

Phone: (413) 781-0070	Project Name: AmherstPelham Regional Middle School/Gale Associat	Analysis Type: PLM
Results to: Derrick.wissman@oneatlas.com	Project Address: 170 Chestnut St. Amherst, Ma.	Positive Stop: Yes
Verbal Results: Yes No	Project Manager: Derrick Wissman	Turnaround Time: 7 Days
Cell or Pager 1-413-664-6687	Project Number: 183DW22062	Fax Copy by:
Site Fax:	Sampled by: Jim Lowell	Return Samples: Yes
Additional Fax:	Special Instructions	No

Lab ID	Field ID	Location	Sample Description	Homogenous Area #
	AMR-09b	A-2 Perimeter Wall Edge	1/8" White Membrane	9
	AMR-09c	B-1 Perimeter Wall Edge	1/8" White Membrane	9
	AMR-10a	A-1 Perimeter Wall Edge	Yellow Mastic Under 1/8" White Membrane	10
	AMR-10b	A-2 Perimeter Wall Edge	Yellow Mastic Under 1/8" White Membrane	10
	AMR-10c	B-1 Perimeter Wall Edge	Yellow Mastic Under 1/8" White Membrane	10
	AMR-11a	A-1 CMU Wall	Clear Caulk on CMU Wall	11
	AMR-11b	A-1 CMU Wall	Clear Caulk on CMU Wall	11
	AMR-12a	A-1 Lower Roof	Yellow Mastic on Roof Clips	12
	AMR-12b	A-2 Lower Roof	Yellow Mastic on Roof Clips	12
	AMR-13a	A-1 Lower Roof Wall Edge	Black Roof Coating	13
	AMR-13b	A-2 Lower Roof Wall Edge	Black Roof Coating	13
	AMR-13c	B-1 Lower Wall Edge	Black Roof Coating	13
	AMR-14a	A-2 Lower East Side	Black Residue on Metal Corrugated Deck	14
	AMR-14b	A-2 Lower East Side	Black Residue on Metal Corrugated Deck	14

Relinquished by:	<i>Paul Lowell</i>	Date/Time:	06-07-2022
Received by:	<i>Mark</i>	Date/Time:	6/8/22
Relinquished by:		Date/Time:	
Received by:		Date/Time:	
Relinquished by:		Date/Time:	
Received by:		Date/Time:	
Company Name:	Atlas Technical		
Company Address:	73 William Franks Rd		
City:	West Springfield	State:	MA
		Zip:	01089
Phone: (413) 781-0070	Project Name: AmherstPelham Regional Middle School/Gale Associat		Analysis Type: PLM
	Project Address: 170 Chestnut St. Amherst, Ma.		Positive Stop: Yes
Results to: Derrick.wissman@oneatlas.com	Project Manager: Derrick Wissman		Turnaround Time: 7 Days
Verbal Results: Yes No	Project Number: 183DW22062		Fax Copy by:
Cell or Pager 1-413-664-6687	Sampled by: Jim Lowell		Return Samples: Yes
Site Fax:	Special Instructions		No
Additional Fax:			
Lab ID	Field ID	Location	Sample Description
	AMR-15a	A-3 Lower Perimeter Wdge	Black Coat on Wood Perimeter Edge
	AMR-15b	A-3 Lower Perimeter Wdge	Black Coat on Wood Perimeter Edge
	AMR-16a	A-2 Lower Roof	Black Mop Seal on Concrete Slab
	AMR-16b	C-Auditorium on South Side	Black Mop Seal on Concrete Slab
	AMR-17a	B-1 Lower Wall Edge	Brown Caulk on Brick Walls
	AMR-17b	B-2 Lower Edge	Brown Caulk on Brick Walls
	AMR-18a	B-3 Inner Wall Edge	White Caulk on Brick Wall
	AMR-18b	B-3 Inner Wall Edge	White Caulk on Brick Wall
	AMR-19a	C-Auditorium Sides	Black Seam Caulk on Metal Flashing
	AMR-19b	C-Auditorium Sides	Black Seam Caulk on Metal Flashing
	AMR-20a	C-Auditorium Roof	1/8" White Membrane
	AMR-20b	C-Auditorium Roof	1/8" White Membrane
	AMR-21a	C-Auditorium Roof	Yellow Mastic Under 1/8" White Membrane
			Homogenous Area #
			15
			15
			16
			16
			17
			17
			18
			18
			19
			19
			20
			20
			21

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307 West 38th Street
New York, NY 10018
(212) 290-0051

10/10/11

032209137

[illegible]